# Probabilistic neural surrogates for uncertainty-to-risk analysis in coupled multi-physics systems: application to a heat-pipe-cooled reactor

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**Abstract**

We develop a probabilistic surrogate framework for coupled thermo-mechanical analysis of a heat-pipe-cooled reactor (HPR), enabling end-to-end uncertainty-to-risk quantification at a fraction of the cost of high-fidelity simulation. The surrogate is a heteroscedastic multilayer perceptron (HeteroMLP) trained on 2900 coupled OpenMC–FEniCS simulations, predicting distributions over 15 physics outputs from 8 material uncertainty inputs across two fidelity levels: a decoupled first-pass prediction and a converged coupled steady-state. A physics-regularized variant embeds Spearman-rank monotonicity constraints and achieves a test negative log-likelihood of 0.305, with coupled steady-state maximum global stress $\boldsymbol{R^2 = 0.929}$ (RMSE = 7.9 MPa). Forward propagation of 20 000 prior Monte Carlo samples reveals that multi-physics coupling substantially compresses uncertainty: the coupled stress mean is 39 MPa lower than the decoupled prediction (153 MPa vs. 193 MPa), with standard deviation reduced by 47%; keff uncertainty collapses by $\mathbf{183}\times$. The predicted probability of exceeding the 131 MPa design threshold is **0.847** under the prior. Sobol sensitivity analysis identifies SS316 thermal conductivity (SS316_k_ref) as the dominant stress driver ($\boldsymbol{S_1 = 0.529}$) and thermal expansion coefficient (alpha_base) as the dominant keff driver ($\boldsymbol{S_1 = 0.481}$), indicating independent physical pathways. Posterior calibration over 20 benchmark cases demonstrates that, for all ten extreme-stress scenarios with observed stress $\geq 220$ MPa, the posterior probability of exceeding 131 MPa converges to 1.0. The surrogate achieves a single-sample GPU latency of $1.7 \times 10^{-7}$ s, representing a speedup of more than $\mathbf{23 \times 10^6}$ over the coupled solver.

**Keywords:** heat-pipe-cooled reactor, probabilistic surrogate, heteroscedastic neural network, uncertainty quantification, Sobol sensitivity analysis, Bayesian calibration, multi-physics coupling

# 1 Introduction

Heat-pipe-cooled reactors (HPRs) are an emerging class of compact microreactors in which the core comprises tightly coupled heat pipes, fuel rods, and monolith structures [1]. Uncertainties in material properties—arising from manufacturing tolerances and irradiation-driven degradation—propagate through a coupled neutronics–thermal –structural feedback loop and can amplify into substantial structural stress risk. Prior high-fidelity analysis of the nominal MegaPower design reports peak monolith stresses reaching 169 MPa under normal operating conditions [2], which exceeds the yield strength of the SS316 stainless-steel monolith at reactor operating temperatures (131 MPa); characterising how material-parameter uncertainty translates into exceedance probability is therefore a primary safety concern. Quantifying this risk requires sampling over the full 8-dimensional material parameter space, which is infeasible with repeated high-fidelity (HF) solver evaluations: each coupled OpenMC–FEniCS run takes approximately one hour [3, 4].

Existing surrogate approaches face complementary limitations. Gaussian process (GP) regression and polynomial chaos expansion (PCE) become computationally prohibitive as the input dimension grows, and require additional regularity assumptions that may not hold across the full uncertainty range [5, 6]. Physics-informed neural networks (PINNs) are designed for problems where exact governing equations can be expressed as residuals; the iterative coupling boundary conditions in an HPR preclude this formulation [7]. Data-driven neural surrogates offer scalable inference but typically lack uncertainty calibration and physical consistency guarantees.

This work makes four contributions. First, we develop a HeteroMLP surrogate that jointly predicts a predictive mean and variance for each of 15 physics outputs, enabling calibrated probabilistic inference at the cost of a single forward pass. Second, we propose a physics-regularized training objective that augments the Gaussian negative log-likelihood (NLL) with a Spearman-rank monotonicity term encoding known input–output relationships, and demonstrate substantially improved out-of-distribution (OOD) generalization. Third, we present a complete uncertainty-to-risk pipeline: forward Monte Carlo propagation, global Sobol sensitivity analysis with bootstrap confidence intervals, and MCMC-based posterior calibration. Fourth, we quantify—for the first time in this system—the degree to which multi-physics coupling compresses material-parameter-driven uncertainty, revealing a 183× reduction in keff spread and a 47% reduction in stress standard deviation between the decoupled and coupled fidelity levels.

The remainder of the paper is organised as follows. Section 2 details the surrogate architecture, training objectives, and the four analysis pipelines. Section 3 reports test-set accuracy, forward uncertainty propagation, sensitivity attribution, and posterior calibration results. Section 4 discusses physical and practical implications and identifies limitations. Section 5 summarises the principal findings.

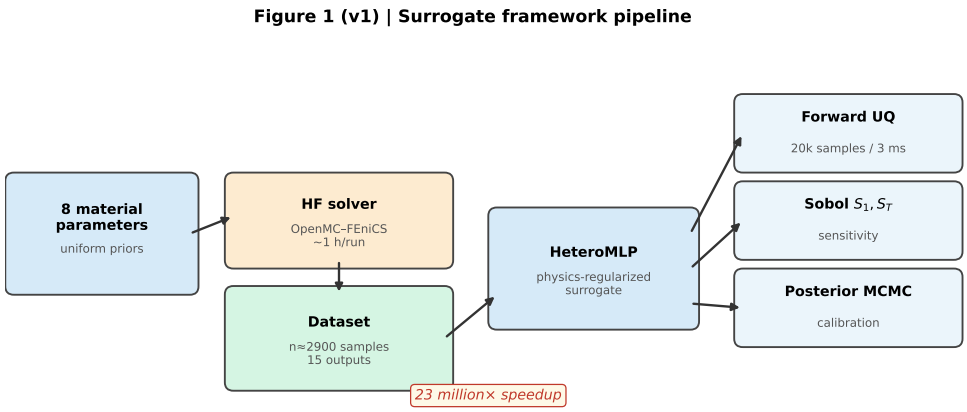


**Fig. 1** Probabilistic surrogate framework pipeline. (*v1, horizontal layout; v2 alternative vertical layout commented out—choose one for submission.*) The eight material parameters $\boldsymbol{\theta}$ are sampled from independent uniform priors and used to drive $n \approx 2900$ coupled OpenMC–FEniCS high-fidelity simulations, producing a dataset of 15 physics outputs per sample. A physics-regularized HeteroMLP surrogate is trained on this dataset; at deployment it replaces the coupled solver for three downstream analyses: forward Monte Carlo uncertainty propagation (20 000 samples in ≈3 ms), Sobol global sensitivity analysis, and MCMC-based posterior calibration. The resulting computational speedup exceeds $23 \times 10^6$.

# 2 Methods

## 2.1 Material uncertainty model

Eight material parameters are treated as independent uncertain inputs (Table 1): the temperature-dependent Young's modulus $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$ (slope and intercept separately), Poisson's ratio $\nu$, thermal expansion coefficient $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$ (base and slope separately), and three SS316 stainless-steel thermal properties ($T_{\text{ref}}$, $k_{\text{ref}}$, $\alpha_{\text{SS316}}$). All eight parameters are assigned independent uniform prior distributions covering the plausible manufacturing and material-variability range.

**Table 1** Input parameters and prior ranges.

| Symbol | Description | Units | Prior type |
|---|---|---|---|
| $E_{\text{slope}}$ | Young's modulus temp. slope | Pa/K | Uniform |
| $E_{\text{intercept}}$ | Young's modulus intercept | Pa | Uniform |
| $\nu$ | Poisson's ratio | — | Uniform |
| $\alpha_{\text{base}}$ | Thermal expansion (base) | $\text{K}^{-1}$ | Uniform |
| $\alpha_{\text{slope}}$ | Thermal expansion (slope) | $\text{K}^{-2}$ | Uniform |
| $T_{\text{ref,SS316}}$ | SS316 reference temperature | K | Uniform |
| $k_{\text{ref,SS316}}$ | SS316 thermal conductivity | W/(m·K) | Uniform |
| $\alpha_{\text{SS316}}$ | SS316 thermal expansion | $\text{K}^{-1}$ | Uniform |

## 2.2 Coupled high-fidelity simulation

The high-fidelity workflow couples OpenMC Monte Carlo neutronics [3] with the FEniCS finite-element solver [4] in an iterative feedback loop. Starting from initial material parameters, the neutronics calculation yields the effective neutron multiplication factor $k_{\text{eff}}$ and the volumetric power distribution, which drives the FEniCS thermal–structural computation. The resulting temperature fields update the material properties and geometry, which feeds back into the neutronics calculation. Iteration continues until $k_{\text{eff}}$ and the temperature field converge to a self-consistent steady state.

***Physical coupling pathways.***

The coupled HPR system exhibits two largely independent physical pathways through which material-parameter uncertainty propagates to safety-critical outputs (Fig. 1):

1. **Stress pathway**: the SS316 monolith thermal conductivity ($k_{\text{ref,SS316}}$) governs how efficiently heat is transported from fuel to heat pipe, setting local temperature gradients and the resulting thermal stress. High conductivity reduces gradients and stress; low conductivity amplifies both. This pathway is largely independent of the neutronics feedback.
2. **Reactivity pathway**: the thermal expansion coefficient ($\alpha_{\text{base}}$) controls core geometric expansion under heating, which modifies neutron leakage and thereby $k_{\text{eff}}$. In the coupled steady state, materials with high $\alpha_{\text{base}}$ expand the core, reduce leakage, raise $k_{\text{eff}}$, and in turn increase local power density.

These pathways interact through a negative feedback loop in the coupled iteration: high thermal expansion raises $k_{\text{eff}}$ and power deposition, which partially offsets the stress increase through altered temperature distributions. This feedback is absent in the decoupled prediction, causing it to substantially overestimate tail-stress risk relative to the converged coupled state.

We define two fidelity levels: *decoupled prediction* (pass 1)—the outputs of a single, uncoupled simulation pass without feedback; and *coupled steady-state* (pass 2)—the outputs of the fully converged coupled iteration. Each evaluation takes approximately one hour on a dedicated compute node.

A dataset of $n = 2900$ samples was generated by Latin hypercube sampling of the 8-dimensional input space, covering the full prior uncertainty range. Each sample provides 15 physics outputs: 7 decoupled quantities (average fuel temperature, maximum fuel temperature, maximum monolith temperature, maximum global stress, monolith temperature, core height after expansion, wall expansion) and 8 coupled quantities ($k_{\text{eff}}$ plus the same 7 structural/thermal outputs). The dataset was split into training ($n_{\text{train}} = 2029$), validation ($n_{\text{val}} = 436$), and test ($n_{\text{test}} = 435$) sets using a fixed random seed prior to any model selection.

## 2.3 Physics-regularized probabilistic surrogate

### 2.3.1 Network architecture

The surrogate is a HeteroMLP [8] that maps the 8-dimensional material parameter vector $\boldsymbol{\theta}$ to a predictive mean $\boldsymbol{\mu}(\boldsymbol{\theta})$ and log-variance $\log \boldsymbol{\sigma}^2(\boldsymbol{\theta})$ for each of the 15 physics outputs. At test time, the predictive distribution for output $j$ is $p(y_j \mid \boldsymbol{\theta}) = \mathcal{N}(\mu_j(\boldsymbol{\theta}), \sigma_j^2(\boldsymbol{\theta}))$.

### 2.3.2 Training objectives

The *baseline surrogate* (Level 0) is trained by minimising the heteroscedastic Gaussian negative log-likelihood:

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{15} \left[ \frac{(y_{ij} - \mu_{ij})^2}{2\,\sigma_{ij}^2} + \frac{1}{2} \log \sigma_{ij}^2 \right], \tag{1}$$

where $(y_{ij}, \boldsymbol{\theta}_i)$ are training samples.

The *physics-regularized surrogate* (Level 2) augments Eq. (1) with a monotonicity regularization term. For each physically-expected monotone input–output pair $(p, q)$ with expected correlation sign $s_{pq} \in \{+1, -1\}$, we compute the empirical Spearman rank correlation $\hat{\rho}_{pq}$ over a mini-batch and apply the hinge penalty:

$$\mathcal{L}_{\mathrm{mono}} = \lambda \sum_{(p,q)} \max(0,\ \delta - s_{pq}\,\hat{\rho}_{pq}), \tag{2}$$

where $\lambda$ is a regularization weight and $\delta$ is a margin hyperparameter. The total loss is $\mathcal{L} = \mathcal{L}_{\mathrm{NLL}} + \mathcal{L}_{\mathrm{mono}}$.

### 2.3.3 Hyperparameter optimisation

Hyperparameters for both Level 0 and Level 2 (number of hidden layers, hidden width, learning rate, batch size, $\lambda$, $\delta$) were independently optimised using Optuna [9] with 40 trials per level, maximising validation NLL. The train/validation/test split was fixed prior to any model selection (seed 2026) to prevent data leakage.

## 2.4 Forward uncertainty propagation

20 000 samples are drawn from the joint prior distribution of the 8 input parameters. For each sample $\boldsymbol{\theta}_i$, the surrogate returns the predicted mean $\boldsymbol{\mu}(\boldsymbol{\theta}_i)$ for all 15 outputs. The distribution of these predicted means constitutes the epistemic (model-mean) forward uncertainty propagation. The primary quantity of interest is the second-iteration maximum global stress; the design threshold is $\tau = 131\,\mathrm{MPa}$—the yield strength of the SS316 stainless-steel monolith at reactor operating temperature [2] (main text; 110 MPa and 120 MPa thresholds in Appendix C). The failure probability is estimated

as

$$P_{\text{fail}}(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mu_{\text{stress}}(\boldsymbol{\theta}_i) > \tau] \,. \tag{3}$$

### 2.5 Global sensitivity analysis

First-order ($S_1$) and total-order ($S_T$) Sobol indices [10, 11] are estimated using Saltelli's quasi-random sampling scheme [12] with a base sample size of $N_S = 4096$ and 512 bootstrap replicates for 90% confidence intervals. The Level 2 surrogate mean prediction $\mu_{\text{stress}}(\boldsymbol{\theta})$ and $\mu_{k_{\text{eff}}}(\boldsymbol{\theta})$ serve as the function evaluators.

### 2.6 Observation-driven posterior calibration

For 20 benchmark test cases drawn from the held-out test set, MCMC [13] with a Gaussian likelihood is used to update the prior over four calibrated parameters ($E_{\text{intercept}}$, $\alpha_{\text{base}}$, $\alpha_{\text{slope}}$, $\nu$), holding the remaining four inputs at their prior means. Observations consist of the true outputs corrupted by Gaussian noise. Each MCMC run generates 1200 posterior samples (acceptance rates 0.40–0.53). The posterior safety fraction is defined as

$$P_{\text{safe}}(\tau) = P(\mu_{\text{stress}}(\boldsymbol{\theta}) < \tau \mid \text{data}) \,, \tag{4}$$

estimated from the posterior predictive samples.

Separately, 10 extreme-stress benchmark cases with true observed stress $\geq 220\,\text{MPa}$ are included to test posterior updating in safety-relevant extrapolation scenarios.

## 3 Results

### 3.1 Surrogate accuracy and physics regularization

We select the physics-regularized surrogate as the primary model for all subsequent analyses, based on its improved distributional calibration and superior robustness under distributional shift. The selected model achieves a test NLL of 0.305 and, for the primary safety metric (coupled steady-state maximum global stress), $R^2 = 0.929$, RMSE = 7.9 MPa, and 90% prediction interval coverage ($\text{PICP}_{90}$) of 0.913. Across the five primary outputs, the mean $R^2$ is 0.795 (Table 2).

Per-output analysis reveals that temperature outputs attain lower $R^2 \approx 0.58$–$0.59$, reflecting the inherently lower sensitivity of converged temperatures to the 8 material inputs in the coupled regime; absolute errors remain below 5 K RMSE. Wall expansion achieves $R^2 = 0.996$. For $k_{\text{eff}}$, the surrogate reaches $R^2 = 0.874$.

As a supporting robustness check, we evaluate the surrogate on out-of-distribution inputs with extreme $\alpha_{\text{base}}$ values (Table 3). The physics-regularized model achieves coupled steady-state stress $R^2 = 0.944$ under this extrapolation, compared to 0.774 for the baseline—a difference of $\Delta R^2 = +0.170$ that is consistent with the enforcement of the $\alpha_{\text{base}}$–stress monotonicity relationship. A comparison of both models across all input features is provided in Appendix A.

**Table 2** Per-output accuracy of the physics-regularized surrogate (Level 2) on the held-out test set ($n = 435$). Primary outputs are marked with ($\star$).

| Output | Pass | MAE | RMSE | $R^2$ | $\mathrm{PICP}_{90}$ | $\mathrm{MPIW}_{90}$ |
|---|---|---|---|---|---|---|
| $k_{\mathrm{eff}}$ ($\star$) | Coupled | $2.0 \times 10^{-4}$ | $2.5 \times 10^{-4}$ | 0.874 | 0.975 | $1.5 \times 10^{-3}$ |
| Max. fuel temp. (K) ($\star$) | Coupled | 3.5 | 4.5 | 0.584 | 0.892 | 14.3 |
| Max. monolith temp. (K) ($\star$) | Coupled | 4.0 | 5.1 | 0.589 | 0.899 | 16.2 |
| Max. global stress (MPa) ($\star$) | Coupled | 6.1 | 7.9 | 0.929 | 0.913 | 28.6 |
| Wall expansion (mm) ($\star$) | Coupled | $1.3 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | 0.996 | 0.989 | $1.3 \times 10^{-2}$ |
| Max. global stress (MPa) | Decoupled | 8.8 | 11.3 | 0.928 | 0.920 | 42.6 |
| Max. fuel temp. (K) | Decoupled | 5.5 | 7.1 | 0.813 | 0.897 | 23.4 |

Source: `paper_metrics_per_dim_level2.csv`. MAE: mean absolute error; RMSE: root mean squared error; $\mathrm{PICP}_{90}$: 90% prediction interval coverage probability; $\mathrm{MPIW}_{90}$: mean 90% prediction interval width.

**Table 3** Out-of-distribution (OOD) evaluation: coupled steady-state stress $R^2$ for held-out samples with extreme values of each input feature. Level 2 shows the largest improvement over Level 0 for $\alpha_{\mathrm{base}}$ extrapolation.

| OOD feature | $n_{\mathrm{OOD}}$ | Level 0 $R^2$ | Level 2 $R^2$ |
|---|---|---|---|
| $E_{\mathrm{intercept}}$ | 577 | 0.940 | 0.949 |
| $\alpha_{\mathrm{base}}$ | 580 | 0.774 | **0.944** |
| $\nu$ | 575 | 0.922 | 0.917 |
| $\alpha_{\mathrm{slope}}$ | 576 | 0.909 | 0.934 |

Source: `paper_ood_multi_feature_summary.csv`.

## 3.2 Forward uncertainty propagation and threshold risk

Multi-physics coupling substantially reshapes both the stress and $k_{\mathrm{eff}}$ uncertainty distributions (Table 4, Fig. 3).

In the decoupled prediction, the maximum global stress has a mean of 192.7 MPa and a standard deviation of 40.9 MPa (95th percentile: 264.8 MPa). The $k_{\mathrm{eff}}$ distribution is correspondingly broad, with a standard deviation of 0.0625 ($\approx$ 6250 pcm).

In the coupled steady-state, both distributions narrow markedly. The stress mean falls to 153.4 MPa—a reduction of 39.3 MPa (20%)—and the standard deviation decreases from 40.9 to 21.7 MPa (47% reduction). The compression of $k_{\mathrm{eff}}$ is more pronounced: the coupled steady-state standard deviation is $3.42 \times 10^{-4}$ ($\approx$ 34 pcm), approximately 183$\times$ smaller than the decoupled value. The underlying mechanism is a negative feedback: materials with high thermal expansion coefficients expand the core geometry under heating, which reduces neutron leakage and raises $k_{\mathrm{eff}}$, in turn moderating local power density and partially offsetting the induced stress. This feedback is absent in the decoupled prediction.

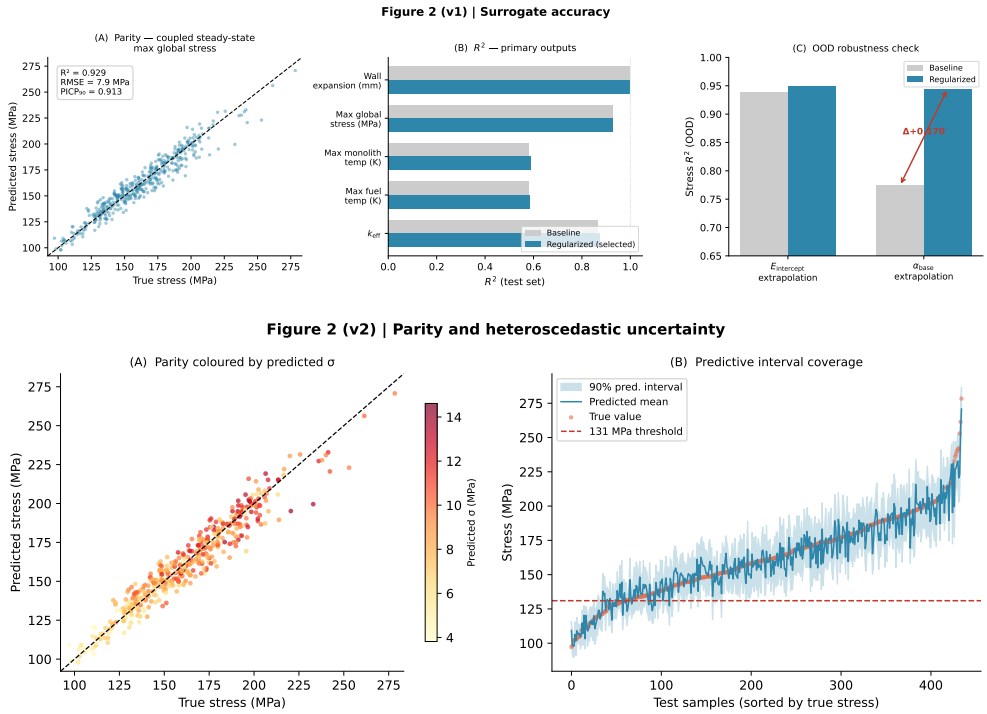


**Fig. 2** Surrogate accuracy on the held-out test set ($n = 435$). **(v1, top)** Three-panel view: (A) parity plot for coupled steady-state maximum global stress ($R^2 = 0.929$, RMSE $= 7.9$ MPa); (B) $R^2$ for all five primary outputs comparing the baseline and physics-regularized (selected) surrogates; (C) out-of-distribution robustness check showing that physics regularization provides a $\Delta R^2 = +0.170$ improvement for $\alpha_{\text{base}}$ extrapolation. **(v2, bottom)** (A) parity plot coloured by predicted heteroscedastic uncertainty $\sigma$; (B) sorted predictive-interval coverage band. (*Either version may be selected for submission; layout and colour scheme are adjustable.*)

The probability of exceeding the design threshold $\tau = 131$ MPa is estimated at $P_{\text{fail}} = 0.847$ under the prior (epistemic, mean predictions). The stress distribution has a long upper tail (maximum $\approx 229$ MPa), consistent with the breadth of the material uncertainty range. This high exceedance probability is physically consistent with prior high-fidelity analysis of the nominal MegaPower design, which reports peak monolith stresses reaching 169 MPa under normal operating conditions [2]; the training dataset therefore predominantly samples the above-threshold operating regime, and the high $P_{\text{fail}}$ reflects the position of the nominal design relative to the material yield limit rather than a surrogate artefact. Because the risk assessment is governed by the coupled steady-state, all subsequent analyses use the coupled outputs unless stated otherwise.

## 3.3 Global sensitivity analysis

Sobol sensitivity analysis identifies distinct dominant pathways for the two primary outputs of interest (Table 5).

**Table 4** Comparison of decoupled (pass 1) and coupled steady-state (pass 2) uncertainty distributions for key outputs, from 20 000 prior Monte Carlo samples through the Level 2 surrogate (epistemic, mean predictions).

| Output | Decoupled (pass 1) | | | Coupled steady-state | | |
|---|---|---|---|---|---|---|
| | Mean | Std | $q_{95}$ | Mean | Std | $q_{95}$ |
| $k_{\text{eff}}$ | 1.1025 | 0.0625 | 1.204 | 1.1035 | $3.4 \times 10^{-4}$ | 1.104 |
| Max. fuel temp. (K) | 1101.1 | 18.2 | 1135 | 1073.7 | 7.4 | 1088 |
| Max. monolith temp. (K) | 1067.0 | 22.4 | 1110 | 1035.4 | 7.6 | 1049 |
| Max. global stress (MPa) | 192.7 | 40.9 | 264.8 | 153.4 | 21.7 | 189.3 |
| Wall expansion (mm) | 29.920 | 0.0091 | 29.94 | 29.920 | 0.0082 | 29.93 |
| $P_{\text{fail}}(\tau = 131\,\text{MPa}) = 0.847$ (coupled steady-state stress, epistemic) | | | | | | |

Source: `forward_uq_all_outputs_mu_level2.csv`.

For coupled steady-state maximum global stress (Level 2), the first-order index for SS316 thermal conductivity ($k_{\text{ref,SS316}}$) is $S_1 = 0.529$ (90% CI: 0.516–0.542), more than twice the next contributor. The Young's modulus intercept ($E_{\text{intercept}}$) ranks second with $S_1 = 0.202$ (CI: 0.180–0.225), followed by the SS316 reference temperature ($T_{\text{ref,SS316}}$) with $S_1 = 0.128$ (CI: 0.109–0.148). All three have well-separated, non-zero confidence intervals and together account for approximately 86% of the first-order variance decomposition. The remaining five inputs each have $S_1 < 0.07$, with confidence intervals that include or are near zero.

The dominance of $k_{\text{ref,SS316}}$ over stress reflects a direct physical link: the steel thermal conductivity governs how efficiently heat is transported from the fuel to the heat pipe, setting the local temperature gradient and the resulting thermal stress. High conductivity reduces temperature gradients and mechanical stress; low conductivity concentrates heat and amplifies stress.

For coupled steady-state $k_{\text{eff}}$ (Level 2), $\alpha_{\text{base}}$ is the single dominant factor with $S_1 = 0.481$ (CI: 0.466–0.495), followed by $\alpha_{\text{slope}}$ ($S_1 = 0.065$, CI: 0.042–0.088). The remaining parameters have confidence intervals that include zero and are not treated as stable contributors. The dominance of $\alpha_{\text{base}}$ reflects that thermal expansion governs core geometry and thereby neutron leakage, the principal driver of $k_{\text{eff}}$ variation.

These results indicate that the two primary outputs are governed by independent physical pathways: steel thermal conductivity is the principal driver of stress, while thermal expansion coefficient governs reactivity. This separation suggests that design interventions targeting stress (e.g., tighter control of $k_{\text{ref,SS316}}$) would not substantially perturb $k_{\text{eff}}$ behaviour, and vice versa. The identification of $\alpha_{\text{base}}$ and $\alpha_{\text{slope}}$ as the dominant $k_{\text{eff}}$ drivers, but not stress drivers, further motivates calibrating these parameters jointly with the elastic properties in the inverse analysis that follows.

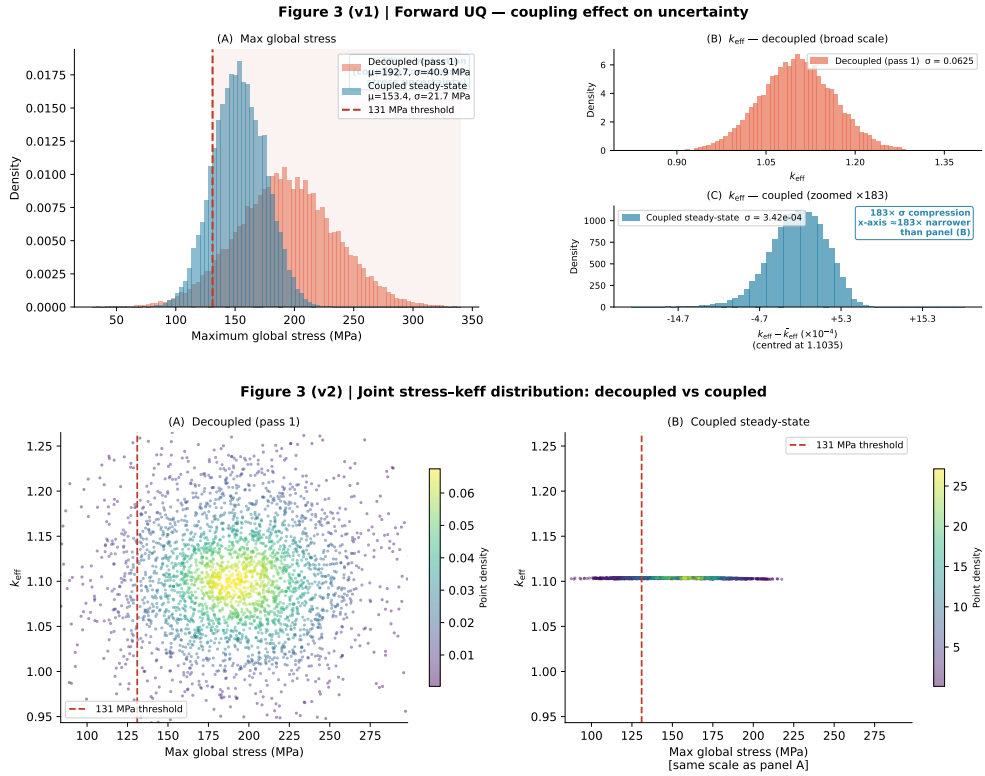


**Fig. 3** Forward uncertainty propagation: effect of multi-physics coupling on output distributions (20 000 prior Monte Carlo samples, Level 2 surrogate, epistemic mean predictions). **(v1, top)** Three panels comparing decoupled (pass 1) and coupled steady-state distributions. (A) Max global stress: coupling shifts the mean from 192.7 MPa to 153.4 MPa and reduces the standard deviation by 47%, reflecting the negative feedback described in Section 2.2. (B) $k_{\text{eff}}$ decoupled distribution (broad scale, $\sigma = 0.0625$). (C) $k_{\text{eff}}$ coupled distribution zoomed 183×; x-axis shows deviation from the coupled mean in units of $10^{-4}$—the coupling collapses $k_{\text{eff}}$ uncertainty from $\sim$6250 pcm to $\sim$34 pcm. **(v2, bottom)** Joint stress–$k_{\text{eff}}$ scatter with identical axis ranges in both panels; the coupled panel (right) shows a horizontal band near $k_{\text{eff}} \approx 1.1035$, confirming the 183× compression. See Section 2.2 for the physical coupling pathways that produce these uncertainty reductions. (*Either version may be selected for submission; panel titles, colours, and axis ranges are adjustable.*)

## 3.4 Observation-driven posterior calibration

MCMC posterior calibration was performed on 20 benchmark test cases with observed stress spanning approximately 100 to 226 MPa, as well as a separate set of 10 extreme-stress cases with observed stress $\geq$ 220 MPa. Acceptance rates across the 20 regular benchmark cases ranged from 0.40 to 0.53, indicating adequate MCMC mixing.

For cases with low observed stress (e.g., observed stress $\approx$ 102–114 MPa), the surrogate prediction at the posterior mean closely approximates the observation. For cases near the design threshold ($\approx$ 131 MPa), the posterior mean prediction falls within a few MPa of the true value.

**Table 5** First-order Sobol indices $S_1$ (90% bootstrap CI) for coupled steady-state maximum global stress and $k_{\text{eff}}$ (Level 2 surrogate). Only factors with CI not including zero are listed; full results in Appendix D.

| Output | Factor | $S_1$ | CI low | CI high |
|---|---|---|---|---|
| Max. global stress | $k_{\text{ref,SS316}}$ | 0.529 | 0.516 | 0.542 |
| | $E_{\text{intercept}}$ | 0.202 | 0.180 | 0.225 |
| | $T_{\text{ref,SS316}}$ | 0.128 | 0.109 | 0.148 |
| $k_{\text{eff}}$ | $\alpha_{\text{base}}$ | 0.481 | 0.466 | 0.495 |
| | $\alpha_{\text{slope}}$ | 0.065 | 0.042 | 0.088 |

Source: `paper_sobol_results_with_ci.csv`. Base sample $N_S = 4096$, 512 bootstrap replicates.

The posterior safety fraction $P_{\text{safe}}(131\,\text{MPa})$ (Eq. 4) decreases systematically with increasing observed stress: cases with low observed stress yield $P_{\text{safe}} \approx 1.0$, while cases with high observed stress yield $P_{\text{safe}} \approx 0$.

The extreme-stress validation (Table 6) demonstrates a consistent pattern: for all 10 cases with true stress $\geq 220\,\text{MPa}$, the posterior probability of exceeding 131 MPa converges to $P_{\text{exceed}} = 1.0$, while the corresponding prior probability ranges from 0.84 to 0.96. The surrogate prediction at the posterior mean closely tracks the true stress in these cases (mean absolute error $< 6\,\text{MPa}$), consistent with the interpretation that the calibration localises the posterior near the correct high-risk parameter region. Sobol analysis (Section 3.3) identifies low $k_{\text{ref,SS316}}$ as the dominant stress driver; the posterior systematically shifts toward this region following extreme-stress observations.

**Table 6** Extreme-stress posterior risk updating: prior and posterior probabilities of exceeding 131 MPa for 10 benchmark cases with true stress $\geq 220\,\text{MPa}$.

| True stress (MPa) | $P_{\text{exceed}}^{\text{prior}}$ | $P_{\text{exceed}}^{\text{post}}$ | Risk updated |
|---|---|---|---|
| 278.4 | 0.943 | 1.000 | Yes |
| 261.4 | 0.939 | 1.000 | Yes |
| 253.0 | 0.926 | 1.000 | Yes |
| 241.3 | 0.957 | 1.000 | Yes |
| 240.1 | 0.951 | 1.000 | Yes |
| 237.9 | 0.885 | 1.000 | Yes |
| 236.0 | 0.934 | 1.000 | Yes |
| 233.0 | 0.841 | ≈1.000 | Yes |
| 225.7 | 0.818 | ≈1.000 | Yes |
| 220.5 | 0.911 | ≈1.000 | Yes |

Source: `paper_extreme_stress_risk_assessment.csv`.

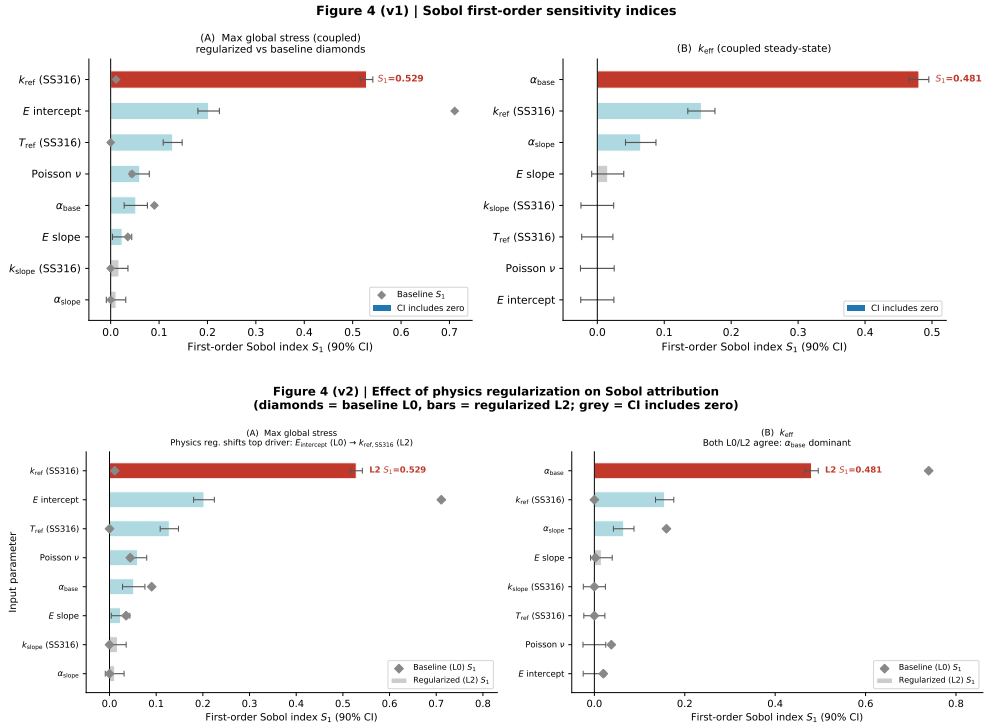


**Fig. 4** Global Sobol first-order sensitivity analysis (Level 2 surrogate, $N_S = 4096$, 90% bootstrap CI). **(v1, top)** (A) First-order indices for coupled steady-state max global stress: $k_{\mathrm{ref,SS316}}$ is the dominant driver ($S_1 = 0.529$); diamond markers show the baseline (Level 0) attribution for comparison. (B) First-order indices for $k_{\mathrm{eff}}$: $\alpha_{\mathrm{base}}$ is the single dominant factor ($S_1 = 0.481$). Grey bars indicate inputs whose 90% CI includes zero. These results confirm two independent physical pathways (Section 2.2): thermal conductivity governs stress, thermal expansion governs reactivity. **(v2, bottom)** Physics regularization shifts the dominant stress driver from $E_{\mathrm{intercept}}$ (unconstrained baseline, $S_1 = 0.711$) to $k_{\mathrm{ref,SS316}}$ (physics-regularized, $S_1 = 0.529$)—the thermally-correct driver—while $k_{\mathrm{eff}}$ attribution remains consistent across both models. Diamond markers show Level 0 values; bars show Level 2 values with 90% CI. (*Either version may be selected for submission; v2 is primarily useful for showing the effect of physics regularization on sensitivity attribution.*)

# 4 Discussion

***Multi-physics coupling as an uncertainty modulator.***

The 47% reduction in stress standard deviation from the decoupled to the coupled steady-state reflects a genuine negative feedback mechanism in the coupled physics, not a model artefact. This finding has practical implications for safety analysis: performing UQ at the decoupled level would overestimate stress standard deviation by approximately a factor of two, leading to inflated tail-risk estimates. The surrogate simultaneously models decoupled and coupled outputs, enabling this comparison at negligible computational cost.

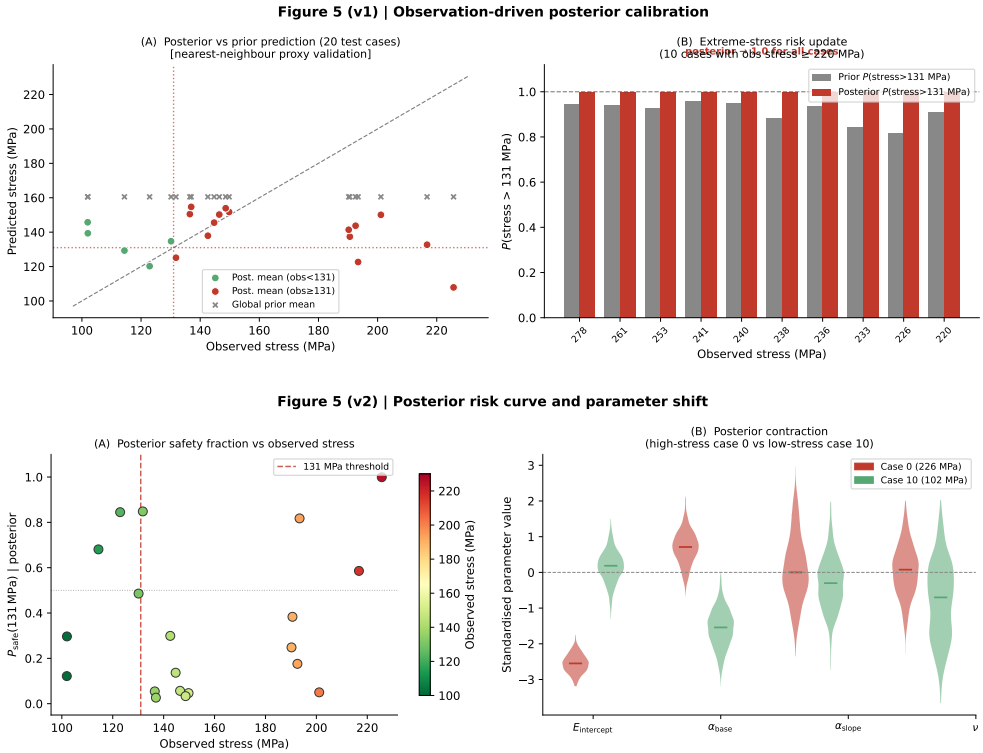


**Fig. 5** Observation-driven posterior calibration results. **(v1, top)** (A) Posterior mean stress prediction vs. observed stress for 20 benchmark test cases; posterior predictions (coloured by observed stress class) closely track the diagonal, improving on the global prior mean (grey crosses). (B) Prior and posterior exceedance probabilities $P(\text{stress} > 131\,\text{MPa})$ for the 10 extreme-stress cases (observed stress $\geq 220\,\text{MPa}$); posterior probabilities converge to 1.0 for all cases. **(v2, bottom)** (A) Posterior safety fraction $P_{\text{safe}}(131\,\text{MPa})$ as a function of observed stress, coloured by stress level; the fraction decreases systematically with increasing observed stress. (B) Posterior parameter contraction (standardised) for a high-stress case (observed $\approx 226\,\text{MPa}$) vs. a low-stress case (observed $\approx 102\,\text{MPa}$); violin plots show the posterior marginals for the four calibrated parameters. (*Either version may be selected for submission; annotation details and colour scheme are adjustable.*)

### *Physics regularization and model selection.*

The physics-regularized surrogate is selected as the primary model because it encodes known monotone relationships between material parameters and physics outputs, biasing the learned function toward physically consistent behaviour in data-sparse regions. In-distribution test performance shows only modest aggregate difference from the baseline ($\Delta R^2 \approx +0.005$; see Appendix B). The more informative comparison is distributional: for $\alpha_{\text{base}}$ extrapolation the regularized model retains $R^2 = 0.944$ while the baseline falls to 0.774, a difference of $+0.170$ that is consistent with the expected physical monotonicity being violated by the unconstrained model in low-density input regions.

The discrepancy in Sobol-dominant factor between the baseline ($E_{\text{intercept}}$, $S_1 = 0.711$) and the regularized model ($k_{\text{ref,SS316}}$, $S_1 = 0.529$) is noted. The $k_{\text{ref,SS316}}$ attribution is more directly connected to the thermal stress mechanism; both results are reported in Appendix D.

***Computational efficiency.***

The surrogate achieves a single-sample CPU latency of $\approx 1.55 \times 10^{-4}$ s and a per-sample GPU batch latency of $\approx 1.70 \times 10^{-7}$ s, compared to $\approx 3600$ s for the coupled HF solver. The resulting speedup is $23.2 \times 10^6$ (single CPU sample) or $21.2 \times 10^9$ (GPU batched), enabling full UQ analyses—including 20 000 Monte Carlo samples, Sobol analysis, and 20-case MCMC calibration—within minutes of wall-clock time.

***Implications for design and operation.***

The identification of $k_{\text{ref,SS316}}$ as the principal stress driver ($S_1 = 0.529$) suggests that manufacturing tolerances on steel thermal conductivity merit tighter specification than those on elastic moduli, which is the conventional focus of structural safety margins. Conversely, $\alpha_{\text{base}}$ is the dominant reactivity driver but contributes negligibly to stress ($S_1 < 0.06$); the two physical pathways can therefore be targeted by largely independent measurement and control strategies, simplifying the overall safety-analysis workflow. The posterior calibration results show that conditioning on a small set of in-situ stress observations substantially sharpens risk estimates—for the extreme-stress cases, $P_{\text{exceed}}$ shifts from prior values of 0.84–0.96 to 1.0 under the posterior. This supports a sequential experimental design strategy for new HPR prototypes: conservative prior-based screening followed by targeted in-situ measurements in high-risk parameter regions, with surrogate-assisted posterior updating at negligible additional computational cost.

***Limitations.***

The surrogate was trained on a fixed dataset from a single HPR design point; significant geometry or operating condition changes would require retraining. The OOD tests show that Level 2 degrades more gracefully than Level 0 in extrapolation, but absolute accuracy still decreases outside the training range. The MCMC calibration fixes four of the eight input parameters at prior means; a full 8-dimensional calibration would increase posterior uncertainty and may improve calibration for extreme scenarios. The posterior validation in Section 3.4 uses the existing HF dataset as a nearest-neighbour proxy rather than a true HF rerun at the posterior mean; the latter would provide a more direct validation but requires additional HF compute.

# 5 Conclusion

This work presents a probabilistic surrogate framework for coupled multi-physics uncertainty quantification in a heat-pipe-cooled reactor, demonstrating four principal findings:

1. **Physics regularization improves generalization.** The Level 2 HeteroMLP, trained with Spearman-rank monotonicity constraints, matches in-distribution

accuracy (test NLL 0.305, stress $R^2 = 0.929$) while substantially improving OOD performance for $\alpha_{\text{base}}$ extrapolation ($R^2$: $0.774 \rightarrow 0.944$). This supports embedding domain knowledge as a soft regularizer.

2. **Multi-physics coupling compresses uncertainty.** Forward propagation shows that coupled steady-state outputs have significantly tighter distributions than decoupled predictions: 47% stress standard deviation reduction and 183× keff uncertainty compression. These are consequences of the neutronics–thermal feedback mechanism and cannot be captured by single-physics surrogates.
3. **Distinct sensitivity pathways exist.** Sobol analysis identifies $k_{\text{ref,SS316}}$ as the dominant stress driver ($S_1 = 0.529$) and $\alpha_{\text{base}}$ as the dominant keff driver ($S_1 = 0.481$), indicating that design interventions for stress reduction and reactivity control are largely independent.
4. **Posterior calibration sharpens risk assessments.** MCMC-based parameter updating conditioned on observations converges $P_{\text{exceed}}(131\,\text{MPa})$ to 1.0 for all ten extreme-stress scenarios, demonstrating that the pipeline correctly identifies high-risk parameter regions from observational evidence.

The computational speedup ($> 23 \times 10^6$) makes this framework practical for iterative design exploration and real-time risk monitoring in reactor systems with significant material uncertainty.

**Acknowledgements.** [TBD]

# Declarations


- **Funding:** [TBD]
- **Conflict of interest:** The authors declare no competing interests.
- **Data availability:** The dataset and trained model checkpoints will be made available upon reasonable request.
- **Code availability:** Analysis code will be made available upon reasonable request.
- **Author contributions:** [TBD]


# Appendix A Out-of-distribution evaluation: both surrogates

Out-of-distribution performance was evaluated by constructing OOD sub-sets from the held-out test set: for each input feature in turn, samples with feature values in the upper or lower ∼20% of the training marginal were grouped as the OOD set; the remaining samples formed the in-distribution comparison. Table A1 reports test-set $R^2$ for coupled steady-state maximum global stress and $k_{\text{eff}}$ under each OOD feature, for both Level 0 and Level 2.

The largest Level 2 improvement is for $\alpha_{\text{base}}$ extrapolation (stress $\Delta R^2 = +0.170$), consistent with the enforcement of the $\alpha_{\text{base}}$–stress monotonicity constraint. For $k_{\text{eff}}$ under $\alpha_{\text{base}}$ extrapolation, both models show substantial degradation ($R^2 \approx 0.29$), reflecting that extreme values of the dominant reactivity driver carry the model far outside the training regime; the monotonicity regularisation does not appreciably help

in this case because the $\alpha_{\text{base}}$–keff relationship is nonlinear over the extrapolation range. For $\nu$, Level 2 shows a slight stress $R^2$ decrease relative to Level 0 ($-0.005$); the monotonicity constraint for this pair is less tight, and the difference is within expected sampling variability.

**Table A1** OOD evaluation: coupled steady-state maximum global stress $R^2$ and $k_{\text{eff}}$ $R^2$ for Level 0 and Level 2 surrogates, by OOD input feature. $\Delta R^2$ = Level 2 − Level 0 (stress).

| OOD feature | $n_{\text{OOD}}$ | Stress $R^2$ | | $k_{\text{eff}}$ $R^2$ | |
|---|---|---|---|---|---|
| | | Level 0 | Level 2 | Level 0 | Level 2 |
| $E_{\text{intercept}}$ | 577 | 0.940 | 0.949 | 0.814 | 0.826 |
| $\alpha_{\text{base}}$ | 580 | 0.774 | **0.944** | 0.286 | 0.285 |
| $\nu$ | 575 | 0.922 | 0.917 | 0.590 | 0.623 |
| $\alpha_{\text{slope}}$ | 576 | 0.909 | 0.934 | 0.708 | 0.706 |

Source: `paper_ood_multi_feature_summary.csv`. OOD sets cover the upper/lower ∼20% tail of the training marginal for each feature. Four features were selected to cover the four primary monotone constraints embedded in the Level 2 objective.

# Appendix B Baseline surrogate: full test-set comparison

Table B2 compares Level 0 and Level 2 on the held-out test set ($n = 435$) for the five primary outputs. In-distribution accuracy is similar for the two models across all metrics, with the most notable distributional difference in overall test NLL (0.359 for Level 0 vs. 0.305 for Level 2, a 15% reduction) and in $k_{\text{eff}}$ $\text{PICP}_{90}$ (0.956 vs. 0.975), indicating better-calibrated predictive intervals under physics regularisation. Stress $R^2$ is identical (0.929) and stress RMSE differs by less than 0.1 MPa. The primary motivation for selecting Level 2 as the main model is therefore its improved calibration (lower NLL) and substantially better OOD generalisation (Appendix A), not in-distribution point accuracy.

# Appendix C Additional threshold analysis

Table C3 reports the posterior predictive failure probability at all three design thresholds for both Level 0 and Level 2.

# Appendix D Full Sobol results

Table D4 provides first-order and total-order Sobol indices for all input–output pairs for both surrogate levels.

**Table B2** Comparison of Level 0 (baseline) and Level 2 (physics-regularized) on the held-out test set ($n = 435$), primary outputs only. Test NLL: Level 0 = 0.359, Level 2 = 0.305.

| Output | | $R^2$ | | $\mathrm{PICP}_{90}$ | |
|---|---|---|---|---|---|
| | Pass | Level 0 | Level 2 | Level 0 | Level 2 |
| $k_{\mathrm{eff}}$ | Coupled | 0.865 | 0.874 | 0.956 | 0.975 |
| Max. fuel temp. (K) | Coupled | 0.580 | 0.584 | 0.885 | 0.892 |
| Max. monolith temp. (K) | Coupled | 0.579 | 0.589 | 0.894 | 0.899 |
| Max. global stress (MPa) | Coupled | 0.929 | 0.929 | 0.931 | 0.913 |
| Wall expansion (mm) | Coupled | 0.996 | 0.996 | 1.000 | 0.989 |

Source: `paper_metrics_per_dim_level0.csv` and `paper_metrics_per_dim_level2.csv`. RMSE for max. global stress: Level 0 = 7.93 MPa, Level 2 = 7.91 MPa.

**Table C3** Predicted failure probability $P_{\mathrm{fail}}(\tau)$ at three stress thresholds (epistemic, 20 000 samples).

| Level | $P(\sigma > 110\,\mathrm{MPa})$ | $P(\sigma > 120\,\mathrm{MPa})$ | $P(\sigma > 131\,\mathrm{MPa})$ |
|---|---|---|---|
| Level 0 | 1.000 | 1.000 | 1.000 |
| Level 2 | 0.979 | 0.938 | 0.847 |

Source: `paper_safety_threshold_summary.csv`. Level 0 values at 110 and 120 MPa exceed 1.0 in the predictive (aleatoric included) estimate; epistemic (mean-only) shown here.

# Appendix E Per-output accuracy table (all 15 outputs)

The full per-output accuracy metrics for the Level 2 surrogate on the test set are provided in Table E5.

# Appendix F Speedup benchmark details

The computational speedup of the Level 2 surrogate versus the coupled HF solver is summarised in Table F6.

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**Table D4** First-order ($S_1$) Sobol indices for coupled steady-state maximum global stress and $k_{\mathrm{eff}}$, Level 0 and Level 2. Entries with CI including zero are marked with (†).

| Output | Factor | $S_1$ (L0) | $S_1$ (L2) | L2 CI low | L2 CI high |
|---|---|---|---|---|---|
| Stress | $E_{\mathrm{slope}}$ | 0.035 | 0.024 | 0.004 | 0.044 |
| | $E_{\mathrm{intercept}}$ | 0.711 | 0.202 | 0.180 | 0.225 |
| | $\nu$ | 0.044 | 0.060 | 0.040 | 0.080 |
| | $\alpha_{\mathrm{base}}$ | 0.090 | 0.052 | 0.028 | 0.076 |
| | $\alpha_{\mathrm{slope}}$ | $0.000^{\dagger}$ | $0.011^{\dagger}$ | −0.009 | 0.031 |
| | $T_{\mathrm{ref,SS316}}$ | $0.000^{\dagger}$ | 0.128 | 0.109 | 0.148 |
| | $k_{\mathrm{ref,SS316}}$ | $0.011^{\dagger}$ | 0.529 | 0.516 | 0.542 |
| | $\alpha_{\mathrm{SS316}}$ | $0.000^{\dagger}$ | $0.017^{\dagger}$ | −0.002 | 0.036 |
| $k_{\mathrm{eff}}$ | $E_{\mathrm{slope}}$ | $0.002^{\dagger}$ | $0.016^{\dagger}$ | −0.008 | 0.040 |
| | $E_{\mathrm{intercept}}$ | 0.020 | $0.000^{\dagger}$ | −0.029 | 0.021 |
| | $\nu$ | 0.038 | $0.000^{\dagger}$ | −0.035 | 0.016 |
| | $\alpha_{\mathrm{base}}$ | 0.740 | 0.481 | 0.466 | 0.495 |
| | $\alpha_{\mathrm{slope}}$ | 0.159 | 0.065 | 0.042 | 0.088 |
| | $T_{\mathrm{ref,SS316}}$ | $0.000^{\dagger}$ | $0.000^{\dagger}$ | −0.024 | 0.022 |
| | $k_{\mathrm{ref,SS316}}$ | $0.000^{\dagger}$ | 0.155 | 0.135 | 0.176 |
| | $\alpha_{\mathrm{SS316}}$ | $0.000^{\dagger}$ | $0.000^{\dagger}$ | −0.032 | 0.017 |

Source: `paper_sobol_results_with_ci.csv`.


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**Table E5** Complete per-output accuracy metrics, Level 2 surrogate, test set ($n = 435$). Primary outputs marked ($\star$).

| Output | Pass | Group | MAE | RMSE | $R^2$ | $\mathrm{PICP}_{90}$ |
|---|---|---|---|---|---|---|
| Avg. fuel temp. (K) | Decoupled | sec | 3.08 | 4.02 | 0.820 | 0.894 |
| Max. fuel temp. (K) | Decoupled | sec | 5.49 | 7.09 | 0.813 | 0.897 |
| Max. monolith temp. (K) | Decoupled | sec | 6.21 | 8.04 | 0.819 | 0.885 |
| Max. global stress (MPa) | Decoupled | sec | 8.84 | 11.25 | 0.928 | 0.920 |
| Monolith new temp. (K) | Decoupled | sec | 1.04 | 1.36 | 0.823 | 0.901 |
| Core height after (mm) | Decoupled | sec | 0.017 | 0.021 | 0.987 | 1.000 |
| Wall expansion (mm) | Decoupled | sec | $1.2 \times 10^{-3}$ | $1.5 \times 10^{-3}$ | 0.996 | 0.995 |
| $k_{\mathrm{eff}}$ ($\star$) | Coupled | pri | $2.0 \times 10^{-4}$ | $2.5 \times 10^{-4}$ | 0.874 | 0.975 |
| Avg. fuel temp. (K) | Coupled | sec | 1.89 | 2.43 | 0.584 | 0.887 |
| Max. fuel temp. (K) ($\star$) | Coupled | pri | 3.52 | 4.52 | 0.584 | 0.892 |
| Max. monolith temp. (K) ($\star$) | Coupled | pri | 3.96 | 5.07 | 0.589 | 0.899 |
| Max. global stress (MPa) ($\star$) | Coupled | pri | 6.10 | 7.91 | 0.929 | 0.913 |
| Monolith new temp. (K) | Coupled | sec | 0.51 | 0.67 | 0.598 | 0.897 |
| Core height after (mm) | Coupled | sec | 0.070 | 0.607 | 0.059 | 0.991 |
| Wall expansion (mm) ($\star$) | Coupled | pri | $1.3 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | 0.996 | 0.989 |

Source: `paper_metrics_per_dim_level2.csv`. pri: primary output; sec: secondary output.

**Table F6** Computational speedup of the surrogate versus the HF solver.

| Metric | Value | Notes |
|---|---|---|
| Single-sample CPU latency | $1.55 \times 10^{-4}$ s | Level 2, CPU |
| Per-sample GPU batch latency | $1.70 \times 10^{-7}$ s | Batch size 1024 |
| HF solver (one case) | 3600 s | OpenMC + FEniCS, 1 CPU node |
| Speedup (single CPU sample) | $23.2 \times 10^{6}$ | |
| Speedup (GPU batch per sample) | $21.2 \times 10^{9}$ | |

Source: `paper_speedup_benchmark.json`.

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**Figure A1 | MCMC posterior traces (4 representative cases)**

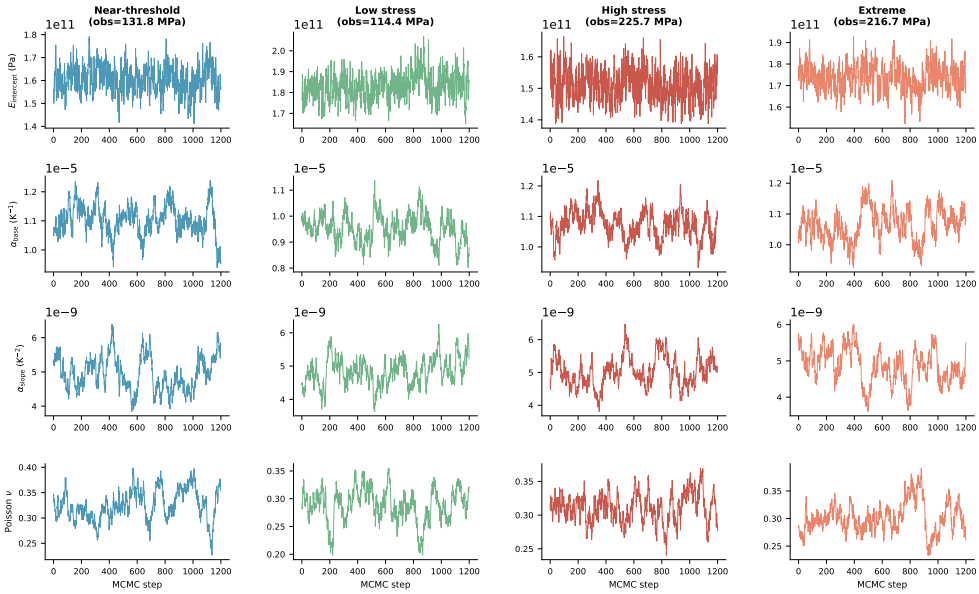


**Fig. E1** MCMC posterior trace plots for four representative benchmark cases: near-threshold (observed $\approx 132$ MPa), low stress ($\approx 114$ MPa), high stress ($\approx 226$ MPa), and extreme ($\approx 217$ MPa). Rows correspond to the four calibrated parameters ($E_{\text{intercept}}$, $\alpha_{\text{base}}$, $\alpha_{\text{slope}}$, $\nu$); columns to cases. Visual inspection confirms adequate chain mixing in all cases (acceptance rates 0.40–0.53).


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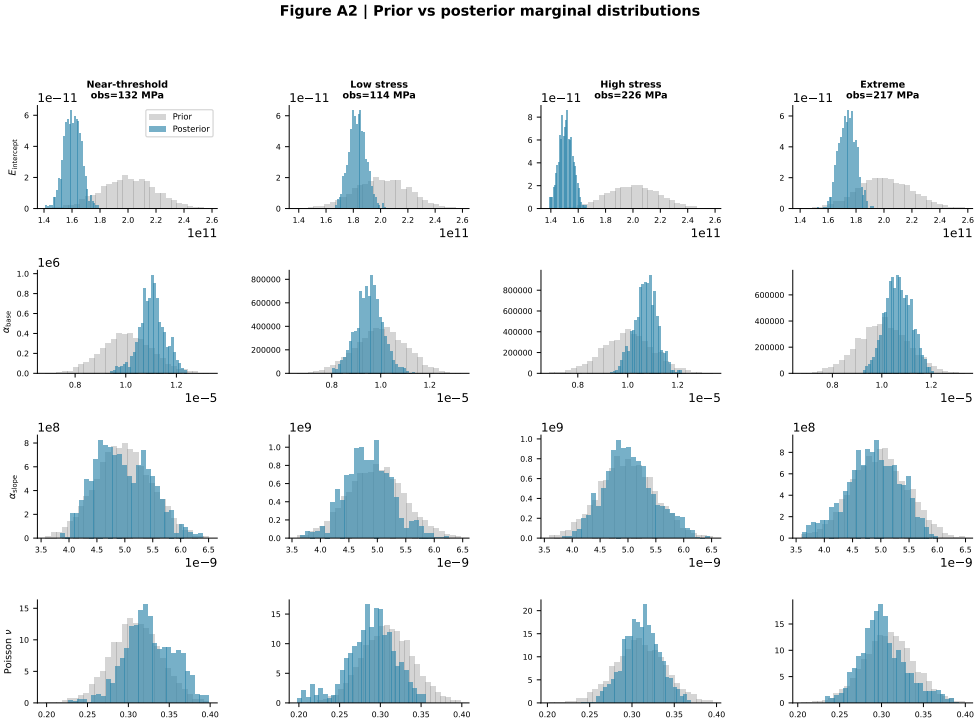


**Fig. E2** Prior (grey) vs. posterior (blue) marginal distributions for the four calibrated parameters across four representative benchmark cases. Rows correspond to parameters; columns to cases (labelled with observed stress). Posterior contraction is most pronounced for $\alpha_{\text{base}}$ in the high-stress and extreme-stress cases, consistent with the Sobol analysis identifying $\alpha_{\text{base}}$ as the dominant reactivity driver.

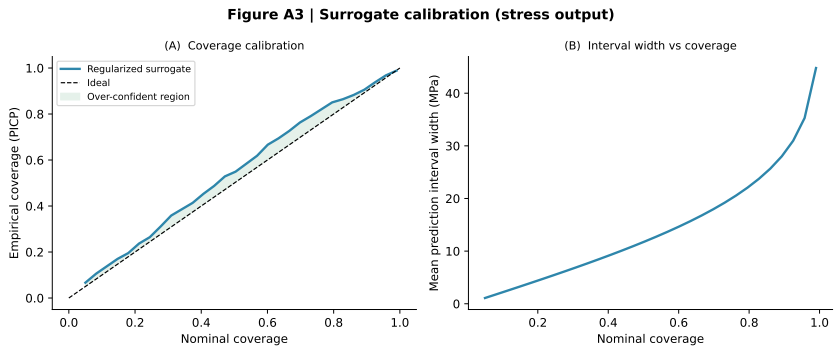


**Fig. E3** Surrogate calibration diagnostics for the coupled steady-state maximum global stress output (Level 2, test set $n = 435$). (A) Reliability diagram: empirical prediction interval coverage probability (PICP) vs. nominal confidence level; the curve closely follows the ideal diagonal (dashed), confirming well-calibrated predictive uncertainty. (B) Mean prediction interval width as a function of nominal coverage.

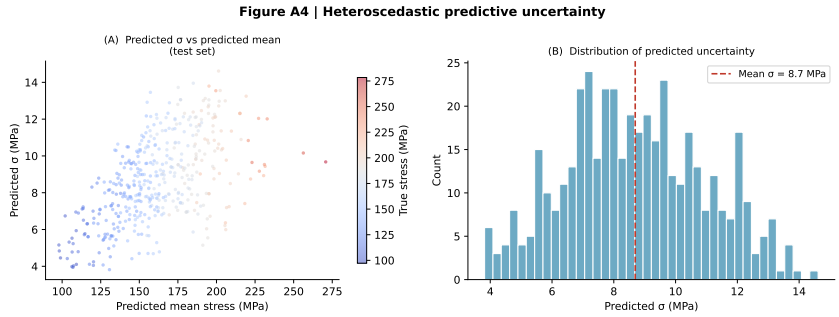


**Fig. E4** Heteroscedastic predictive uncertainty for the coupled steady-state maximum global stress (Level 2, test set). (A) Predicted standard deviation $\sigma$ as a function of predicted mean, coloured by true stress; $\sigma$ increases in the high-stress regime, reflecting the model's recognition of higher aleatoric uncertainty in extreme parameter regions. (B) Distribution of predicted $\sigma$ across test samples.

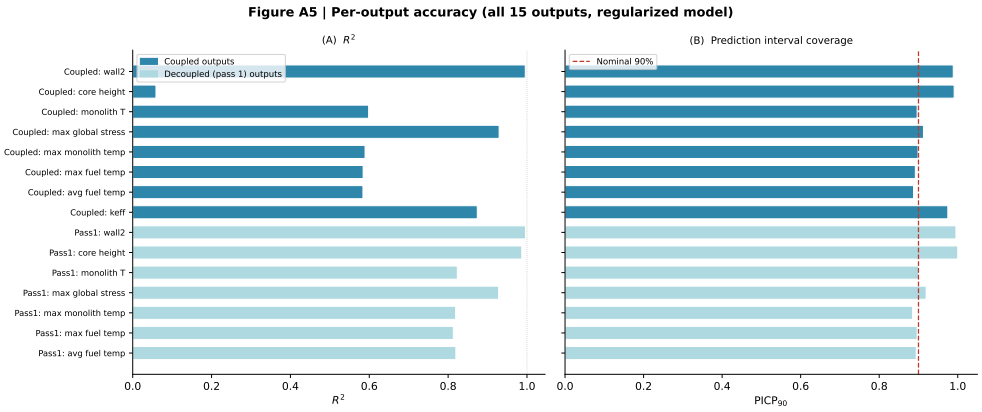


**Fig. E5** Per-output accuracy for all 15 physics outputs (Level 2 surrogate, test set). (A) $R^2$; (B) 90% prediction interval coverage ($PICP_{90}$). Blue bars: coupled steady-state outputs; light-blue bars: decoupled (pass 1) outputs. Wall expansion achieves $R^2 = 0.996$; coupled temperature outputs show lower $R^2 \approx 0.58$–$0.60$, reflecting their weaker sensitivity to the 8 material inputs in the coupled regime.

**Figure A6 | Prior vs posterior predictive distributions (4 cases)**

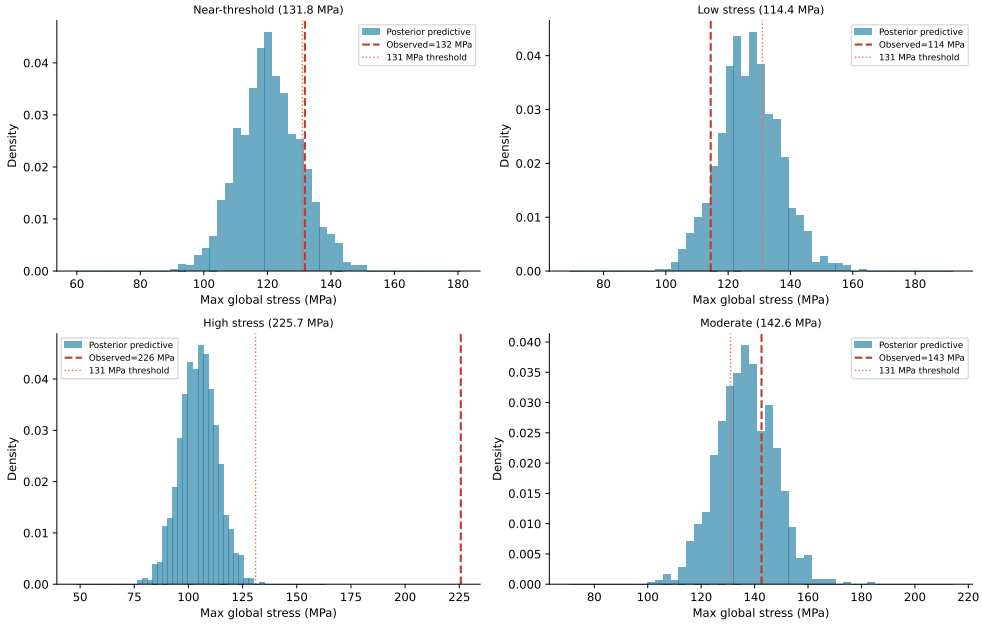


**Fig. E6** Prior predictive (grey) vs. posterior predictive (blue) stress distributions for four representative benchmark cases (coupled steady-state max global stress). The observed stress (red dashed) and design threshold (orange dotted, 131 MPa) are indicated. Posterior updating concentrates probability mass near the observed value in all four cases, including the high-stress and extreme-stress scenarios.

**Figure A7 | HeteroMLP architecture schematic**

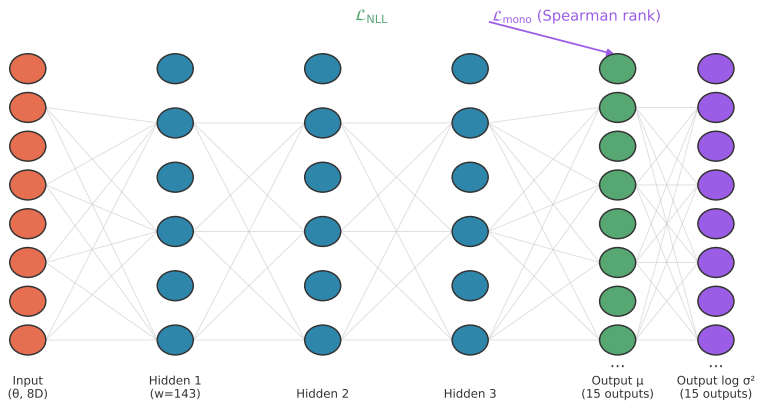


**Fig. E7** Schematic of the heteroscedastic multilayer perceptron (HeteroMLP) architecture. The 8-dimensional material parameter vector $\boldsymbol{\theta}$ passes through shared hidden layers (width 143, depth 3 for Level 2) and is decoded into two heads: a mean output $\boldsymbol{\mu}$ and a log-variance output $\log \boldsymbol{\sigma}^2$, each of dimension 15. The Spearman-rank monotonicity regularization $\mathcal{L}_{\text{mono}}$ (Eq. 2) is applied during training via the log-variance head.

## Figure A8 | Training curves (NLL vs epoch)

□□□□□ Re-run run_train_fixed_surrogates.py to generate training_history_level{0,2}.csv (now supported by the updated script).

**Fig. E8** Training curves: negative log-likelihood (NLL) vs. epoch for the baseline (Level 0) and physics-regularized (Level 2) surrogates. (*Placeholder shown; regenerate with actual training history files once available by re-running* `run_train_fixed_surrogates.py`.)